In [1]:
```
import numpy as np
import pandas as pd

from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.metrics import mean_squared_error, r2_score

# 1. Load the Diabetes dataset
diabetes = load_diabetes(as_frame=True)

X = diabetes.data
y = diabetes.target

print("Dataset shape:", X.shape)
print("\nFirst five rows:")
print(X.head())

print("\nTarget values:")
print(y.head())

# 2. Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

print("\nTraining samples:", X_train.shape[0])
print("Testing samples:", X_test.shape[0])

# 3. Standard Linear Regression
linear_model = Pipeline([
    ("scaler", StandardScaler()),
    ("model", LinearRegression())
])

linear_model.fit(X_train, y_train)
linear_predictions = linear_model.predict(X_test)

linear_mse = mean_squared_error(y_test, linear_predictions)
linear_r2 = r2_score(y_test, linear_predictions)

# 4. Ridge Regression with cross-validation
ridge_pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("model", Ridge())
])

ridge_params = {
    "model__alpha": np.logspace(-3, 3, 20)
}

ridge_search = GridSearchCV(
    estimator=ridge_pipeline,
    param_grid=ridge_params,
```

```
    cv=5,
    scoring="neg_mean_squared_error",
    n_jobs=-1
)

ridge_search.fit(X_train, y_train)

best_ridge = ridge_search.best_estimator_
ridge_predictions = best_ridge.predict(X_test)

ridge_mse = mean_squared_error(y_test, ridge_predictions)
ridge_r2 = r2_score(y_test, ridge_predictions)

# 5. Lasso Regression with cross-validation
lasso_pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("model", Lasso(max_iter=100000))
])

lasso_params = {
    "model__alpha": np.logspace(-3, 2, 30)
}

lasso_search = GridSearchCV(
    estimator=lasso_pipeline,
    param_grid=lasso_params,
    cv=5,
    scoring="neg_mean_squared_error",
    n_jobs=-1
)

lasso_search.fit(X_train, y_train)

best_lasso = lasso_search.best_estimator_
lasso_predictions = best_lasso.predict(X_test)

lasso_mse = mean_squared_error(y_test, lasso_predictions)
lasso_r2 = r2_score(y_test, lasso_predictions)

# 6. Display the best hyperparameters
print("\nBest Ridge alpha:", ridge_search.best_params_["model__alpha"])
print("Best Lasso alpha:", lasso_search.best_params_["model__alpha"])

print("\nBest Ridge CV MSE:", -ridge_search.best_score_)
print("Best Lasso CV MSE:", -lasso_search.best_score_)

# 7. Compare model performance
results = pd.DataFrame({
    "Model": [
        "Linear Regression",
        "Ridge Regression",
        "Lasso Regression"
    ],
    "MSE": [
        linear_mse,
        ridge_mse,
        lasso_mse
    ],
    "R-squared": [
        linear_r2,
```

```
        ridge_r2,
        lasso_r2
    ]
})

print("\nModel Performance Comparison:")
print(results.round(4).to_string(index=False))

# 8. Identify the best model based on test MSE
best_model_name = results.loc[results["MSE"].idxmin(), "Model"]

print("\nBest model based on test MSE:", best_model_name)
```

```
Dataset shape: (442, 10)

First five rows:
        age       sex       bmi        bp        s1        s2        s3  \
0  0.038076  0.050680  0.061696  0.021872 -0.044223 -0.034821 -0.043401
1 -0.001882 -0.044642 -0.051474 -0.026328 -0.008449 -0.019163  0.074412
2  0.085299  0.050680  0.044451 -0.005670 -0.045599 -0.034194 -0.032356
3 -0.089063 -0.044642 -0.011595 -0.036656  0.012191  0.024991 -0.036038
4  0.005383 -0.044642 -0.036385  0.021872  0.003935  0.015596  0.008142

         s4        s5        s6
0 -0.002592  0.019907 -0.017646
1 -0.039493 -0.068332 -0.092204
2 -0.002592  0.002861 -0.025930
3  0.034309  0.022688 -0.009362
4 -0.002592 -0.031988 -0.046641

Target values:
0    151.0
1     75.0
2    141.0
3    206.0
4    135.0
Name: target, dtype: float64

Training samples: 353
Testing samples: 89

Best Ridge alpha: 26.366508987303554
Best Lasso alpha: 1.8873918221350976

Best Ridge CV MSE: 3121.1533586516784
Best Lasso CV MSE: 3127.0841813448797

Model Performance Comparison:
            Model       MSE  R-squared
Linear Regression 2900.1936     0.4526
 Ridge Regression 2864.1356     0.4594
 Lasso Regression 2799.2300     0.4717

Best model based on test MSE: Lasso Regression
```

In [ ]: